

5 (a) Pigeons are birds that belong to the Columbidae family. They are distributed throughout the world.

- The two sex chromosomes in female pigeons are known as the **Z** chromosome and the **W** chromosome.
- Male pigeons have two **Z** chromosomes.
- One of the genes controlling feather colour in pigeons is located on the **Z** chromosome. The **W** chromosome does **not** have this gene.
- This gene has three alleles, each coding for a different feather colour.
- The allele for ash-red feathers is dominant to the allele for blue feathers, which is dominant to the allele for brown feathers.

Using the symbols, complete Figure 5.1 to show how a cross between an ash-red male pigeon and a blue female pigeon can produce a brown offspring.

symbols: Z^A = **Z** chromosome with allele coding for ash-red colour
 Z^{Bl} = **Z** chromosome with allele coding for blue colour
 Z^{Br} = **Z** chromosome with allele coding for brown colour
 W = **W** chromosome with no gene controlling feather colour

parental phenotypes: ash-red male blue female

parental genotypes:

offspring

parental gametes		→			
	genotype		genotype		
	phenotype		phenotype		
	genotype		genotype		
	phenotype		phenotype		

Figure 5.1



(b) State the type of inheritance shown in this cross.

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..... [2]

(c) Describe how you would determine the genotype of an ash-red male pigeon.

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